Modeling wave propagation using cellular automata on Chip

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***Abstract*—Cellular automata (CA) are often used for physical modelings in which space and time are discrete by assumption. An artificial organism can be considered as a circuit that controls a high-level integrated CA architecture, developed for some mathematical purpose. In this work, propagation and reflection wave phenomenon has been reproduced using CA systems, in a one-dimensional lossless medium. The main objective here was to develop a modeling tool, called *vCAgen*, which provides the VHDL code for the CA-based system implementation. The CA-based system called *Camphs1D*, composed of 185 cells and two receiver channels, was mapped on a Xilinx Zynq Ultrascale 7EV1156 device, using 27-bit width floating-point arithmetic representation. The proposed *Camphs1D* was effectively mapped on the Zynq device, achieving an operational frequency of 100 MHz and throughput of 25 MOPS. Numerical comparisons allow us to conclude that the proposed CA-system, which is based on simple arithmetic operations, achieves the same results that a reference model based on the well known d'Alembert one-dimensional discrete wave solution, with a Mean Square Error (MSE) in the order of 10^{-13}**

***Index Terms*—Cellular automata, Wave phenomenon, SoC FPGA.**

I. Introduction

Physical modelings of acoustics waves, electromagnetic waves, fluid dispersion, and its interactions with obstacles in space, commonly make use of partial differential equations (PDEs) [20]. Traditional methods for solving PDEs such as finite difference methods (FDM) require a fine discretization of space for achieving accurate solutions and are very dependent on initial and boundary conditions [16] [18]. For this kind of solution, large computational resources are required, which makes it inefficient for modeling complex PDE systems.

In order to overcome the challenges imposed by continuous space-time models, the cellular automata (CA) can model complex scenarios using simple preconditions over the behavior of spatial and temporally discrete units, in a wide variety of fields. However, there is a problem regarding the computational costs of CA-based systems in cases of huge discrete domains, which results in exhaustive simulations. This issue is a drawback for designing systems that commonly imposes the necessity of solving the inverse problem and then involves hundreds or thousands of simulations of the associated forward model. One possible solution to that problem is the hardware acceleration of the CA-based models, overcoming the problem of repetitive and enormous calculations [1] [9].

This work presents a modeling tool for one-dimensional CA systems called *vCAgen*, which provides the VHDL code, allowing the obtained models to be implemented in system-on-a-chip (SoC) devices, accelerating its execution time. In particular, this work proposes a new CA model for one-dimensional discrete domains, forced by monochromatic harmonic signals, in a lossless medium. Obstacles have been used to model the frontiers and a membrane inside the linear grid, allowing the representation of basic wave phenomenons and their constructive interactions.

A reference software model was built using C language, to reproduce a traditional numerical solution for the partial differential one-dimensional wave equation, in discrete domain as well. The implemented cell's core, for the designed CA system, called *Camphs1D*, makes use of floating point arithmetic IP-cores with custom precision, to hand over all the arithmetical needed processes. It is important to highlight that the custom precision provided by the IP-cores, allow the designer to analyze the trade-off between hardware resources consumption and numerical precision. For a deep study on the floating point IP-cores, readers can refer to the work in [13], [14].

In order to validate the proposed *Camphs1D* solution, a CA system, composed of 185 cells and two source channels, was mapped on a Zynq Ultrascale 7EV1156 device from Xilinx. Behavioral simulations demonstrate the accuracy of the proposed CA-system with respect to the PDE-based model, achieving a mean square error (MSE) of about 10E-13.

II. Reference Model

A plane wave is a kind of a pure mathematical function, and it does not exist in the real world. However, we must consider these wave functions as the first step in any matter of computation. Considering a harmonic wave traveling along positive x-axis [10] [3],

$$\frac{\partial^2 p(x,t))}{\partial x^2} - \frac{1}{c_0^2}\frac{\partial^2 p(x,t))}{\partial t^2} = 0, \tag{1}$$

The well-known d'Alembert solution for plane waves [2] states that a wave quantity, defined by Equation 1, $p(x,t)$, can be

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represented by a superposition of harmonic concurrent patterns at the same point in the space, as follows,

$$p = f(x - c_0 t) + g(c + c_0 t). \tag{2}$$

It is shown in Fig. 1 an aleatory instant of the transient solution obtained from the reference model. It was constituted of one acoustic source and some obstacles, in a lossless or ideal medium. Some harmonic signals are traveling in both directions which are about to hit an obstacle. Note that equations 1 and 2 do not consider reflections due to obstacles, but only wave dispersion and its constructive interactions. To do it, some numerical efforts and additional theories are needed, however, that development is out of the scope of this paper.

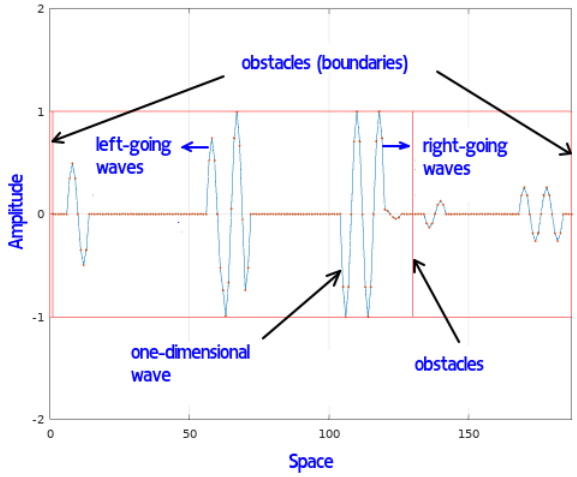


Fig. 1. Simulation of the propagation of an uni-dimensional plane wave.

In this work, several assumptions were made to construct a reference model for a uni-dimensional plane wave and its interactions with the medium and obstacles [7]. The mathematical characteristics of the reference model are shown below.

- Mesh size of 20 meters with 187 cells.
- Velocity of sound wave equal to $344 \sim m/s$.
- Characteristic impedance equal to $400 \sim$ Rayls.
- A membranes with impedance set to $2667 \sim$ Rayls.
- Two membranes at the frontiers of the mesh with impedance set to $1200 \sim$ Rayls.
- Mesh update frequency set to $3200 \sim$Hz.
- Simulation time equal to 0.05 seconds (160 frames).
- Input frequencies equal to $400 \sim$Hz. That means 8 points to plot per period.
- Two sinusoidal inputs at different locations, with 1 and 2 complete periods.

The whole numerical formulation can be found in the references [12], [17], [6]. Sign definitions and algorithms applied to the reference model have been neglected in this work because it is out of scope. In summary, this work uses the d'Alembert solution for plane waves as a reference model, which is computationally solved by an FDM technique, thus allowing for numerical comparisons with the solution achieved by the proposed cellular automata model, explained in the next section.

III. Cellular Automata Modeling

A. Background

In an informal way, but satisfactory, a CA system can be adopted wherever the following characteristics exist [4].

1) It consists of an n-dimensional grid of cells.
2) It evolves over discrete time-steps.
3) Each cell is a kind of finite-state unit whose operation or output values are determined only by its current state and its neighborhood.
4) The neighborhood relations are local and uniform.

In the meantime, the main characteristic of a CA system (one which supports this work) falls within its nature. A CA system is supposed to operate in a truly simple mode, with no geometry restrictions, no trajectories or functions that could stop the spontaneous evolution of each cell.

The multipurpose mathematician Von Neumann was the pioneer in creating a CA system, in the late forties, during his studies over computing machines. The results of this work were published after his death, in 1966, by Arthur W. Burks, [15]. Von Neumann treated his CA systems as a mathematical idealization of natural organisms, including surviving rules, reproducing mechanisms, and temporal evolution. In the seventies, the CA systems became more popular with Conway's "Game of Life" [8]. This simple system needs only three survival rules to show formidable and unexpected results.

Over the years, CA systems have been applied as a versatile modeling tool in different fields of knowledge such as acoustics [19], electromagnetism [11] and fabrication processes [5].

B. CAMPHS1D: Cellular Automata Modeling of One-dimensional Physical Systems

The CA system called *Camphs1D* was designed to represent, as simple as possible, the basic wave phenomenons obtained from the reference model simulation. In terms of its nature, this CA system brings the following operating rules.

1) Wave quantities must be "transmitted" through dispersion cells along both wave guides direction $(+x, -x)$, separately. A dispersion cell does not change its behavior in life.
2) A generating cell must not receive but force wave quantities to the neighborhood, in each discrete time, in both directions. For that, values must be provided to that cell by an external agent, inside the artificial organism. A generating cell might have its behavior changed by the artificial organism to a dispersion cell, any time, either forward or backward.
3) A hybrid cell must receive wave quantities like a dispersion cell and transmit wave quantities like a generating cell. A hybrid cell does not change its behavior in life.

These rules could be materialized through the figures below. These images bring some mathematical relations, according to

TABLE I
CODE PROFILING FOR BOTH STRATEGIES: FDTD AND CA.

Parameter	FDTD solution	CA Solution
Executions number	1000	1000
Time processing	3.36 sec.	3.68 sec.
Multiplication operations	89446	57121
Summation operations	59526	864
Division operations	9	0

the finite differential approaching used in the reference model. The superscript symbols '+' and '-' represent right-going and left-going waves portions, respectively, while the subscribed symbol 'n' represents time.

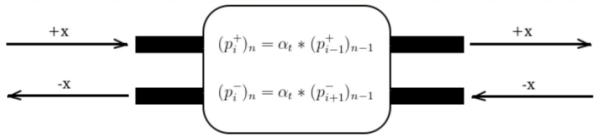


Fig. 2. Rule 1 - Dispersion mechanism.

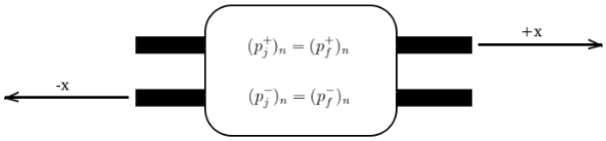


Fig. 3. Rule 2 - Generating mechanism.

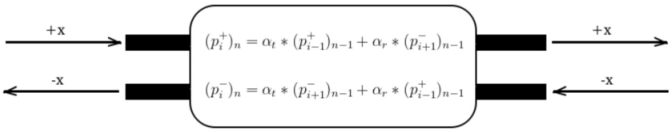


Fig. 4. Rule 3 - Hybrid mechanism.

Just before the VHDL implementation of Camphs1D, a very important and quickly investigation was accomplished, by using of code profiling. Both strategies, FDTD and Cellular automata, had been compared in a software environment. As it can be seen in Table I, the results highlight the CA approaching, pointing out a good opportunity to work with floating point operations, on chip.

Following these rules, during the VHDL description of Camphs1D, it was realized that all of them could be better represented by two components, or two cells. Both cellular automatas, GCA1D and HCA1D, are show in Fig. 5. On one hand, the cellular automata GCA1D has been designed to generate and disperse wave quantities, according to rules 1 and 2. Some of them will never be requested as a generating cell in life because it might not exist any source wave at its position. The GCA1D's core makes use of three floating-point multipliers, two 3×3 multiplexers, two OR gates, and one register. The floating-point operations are supported by the FPUPACK library [13] [14], using a 27-bit width representation. We have also a register bank of 27 bits to store a constant value, normally set to zero. The GCA1D cells work in synchronous mode, with a latency of four clock cycles, due to internal floating-point operations. On the other hand, the automata HCA1D is in charge of wave dispersion at the surfaces of the medium, which means, wave reflections from the obstacles. This cell makes use of four floating-point multipliers and two adders. Because of the same reasons mentioned before, this cell must work with the same latency of four clock cycles. For a more detailed description regarding all signals in Fig. 5, see Table II.

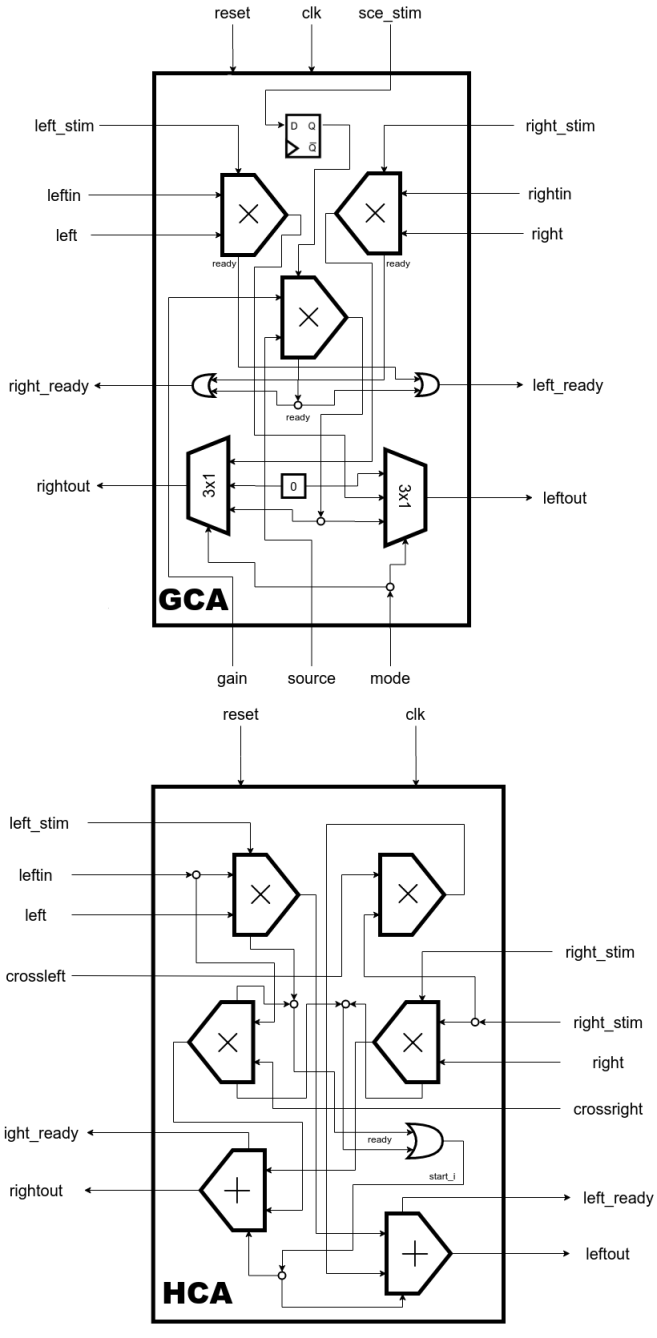


Fig. 5. Operational blocks used in the GCA1D and HCA1D design.

TABLE II
DESCRIPTION OF USED SIGNALS FOR THE GCA1D AND HCA1D CELLS - VCAGEN JOB.

Signal/Gate	IN/OUT	Bits	Description
CLK	IN	1	clock signal
RESET	IN	1	Register/device initialization
MODE	IN	2	Generating/Dispersion/OFF
SOURCE	IN	27	Forced value for sources
GAIN	IN	27	internal gain for sources
SCE_STIM	IN	1	generating stimulus
LEFT_STIM	IN	1	Dispersion stimulus (left)
RIGHT_STIM	IN	1	Dispersion stimulus (right)
LEFT	IN	27	left wave coefficient
RIGHT	IN	27	right wave coefficient
LEFTIN	IN	27	wave quantities from the left
RIGHTIN	IN	27	wave quantities from the right
CROSSLEFT	IN	27	left wave coefficient
CROSSRIGHT	IN	27	right wave coefficient
RIGHT_READY	OUT	1	calculation is done
LEFT_READY	OUT	1	calculation is done
RIGHTOUT	OUT	27	wave quantities to the right
LEFTOUT	OUT	27	wave quantities to the left

IV. CAMPHS1D GENERAL ARCHITECTURE

As stated before, an artificial organism is an integrated and auto-sufficient system, able to control, by using some input-output data, all the cells that compose the organism. An artificial organism can also run for an indeterminate period of time.

The main architecture of the CA-based organism is responsible for the integration of the cells in a functional environment, fully programmed and adjusted to perform the whole computation of the system. It is important to highlight that the main architecture description may be a tough task, because of the high number of cells, components, and routing signals. In order to make it easier and less costly, a modeling tool for one-dimensional CA systems called *vCAgen* was developed in Octave. The *vCAgen* tool is able to describe any typical *Camphs1D* artificial organism, including its sources, receivers, and topology. The *vCAgen* tool is also able to automatically generate the VHDL code that describes the whole CA system. The following list shows the current characteristics of the *vCAgen* tool.

- Control signals definitions.
- Cell to cell interaction, connecting READY signals to its adjacent waveguides.
- Source values attributions according to its respective data banks and MODE signal operation.
- Signal capturing at the GCA1D receiver cells.
- Automatic VHDL component instantiation.

In this work, the *vCAgen* tool was used to create an artificial organism, called *Camphs1DB9G2*, composed of 185 cells and two source channels. Such a configuration is minimally satisfactory to reproduce the reference model in an FPGA chip. The main architecture of the *Camphs1D* is shown in Fig. 6. Note that the cells form two waveguides, each one for a single direction. Several registers are connected to each cell, providing control signals for configuring its expected behavior.

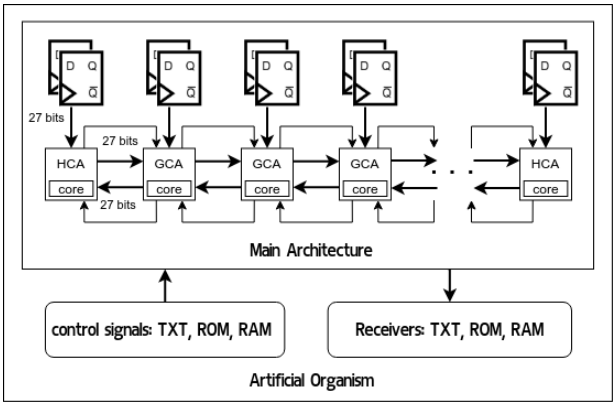


Fig. 6. Operational block used in the design of *Camphs1D* general architecture.

Currently, the *vCAgen* tool provides two ways of validating the behavior of the *Camphs1D* architecture, by injecting the input data and control signals using a TXT file (behavioral simulation) or by using ROMs in case of physical implementation. The signals used by *vCAgen* are shown in II. The output data can be exported to TXT files or can be stored in RAM memory and accessed using the Integrated Logic Analyzer (ILA) core from Xilinx. Eventually, the Virtual Input-Output (VIO) can be used to set inputs and verify outputs.

V. EXPERIMENTAL PROCEDURE

The implemented Camphs1DB9G2 was validated using behavioral simulation. The input data (TXT files) includes *START*, *MODE* and *SOURCE* signals. The output data were collected by the receivers at the 20th and 150th positions, and at every clock cycle, for both directions or waveguides, were written in the respective TXT file.

The Camphs1DB9G2 system has tree HCA1D cells at 2nd, 130th, and 186th positions. It has also two GCA1D cells at 44th and 94th positions, performing two and one complete wave periods, respectively. See Table III for details regarding the generated values. Others cells, namely non-wave-generating and non-hybrid, are GCA1D-type also, but perform only dispersions in both directions, according to their neighborhood stimulus. Two of them have their waves quantities written to TXT files (receivers), in a hexadecimal representation.

The experiment starts with the initialization of the registers, which is done by a *RESET* signal. Then, the CA system is ready to run at any time according to the very first *START* signal. The first generating cell transmits its wave quantities and its *READY* signals, which are connected to the neighborhood, and so on, indefinitely.

The number of lines on each control signal (TXT files) must be synchronized to the system throughput, which is of four clock cycles multiplied by the number of wave discrete events. Thus, the output data needs to be filtered and decoded to be compared to the reference model, in software. Finally, the proposed architecture was characterized in terms of resources

TABLE III
SIMPLE DISCRETE SINUSOIDAL SIGNAL IN TUNE OF 400 ∼HZ BY USING A 3200 ∼HZ UPDATE FREQUENCY. THESE VALUES WERE FORCED AT SOURCE POINTS IN CAMPHS1DB9G2.

Binaries in a 27-bit format	Hexadecimals	400 ∼Hz@3200 ∼Hz
001111110011010100000100111	1F9A827	0.70710563659668
001111111000000000000000000	1FC0000	1
001111110011010100000100111	1F9A827	0.70710563659668
000000000000000000000000000	0000000	0
101111110011010100000100111	5F9A827	-0.70710563659668
101111111000000000000000000	5FC0000	-1
101111110011010100000100111	5F9A827	-0.70710563659668
000000000000000000000000000	0000000	0

consumption, numerical error and precision when compared to the reference model, operational frequency, and execution time (latency and throughput).

VI. RESULTS

All results obtained from the *Camphs1DB9G2* organism were correlated to the reference model. The results provided 160 time frames (each one with 185 values). Figs. 7 and 8 present the obtained CA results in the same plot with the reference model.

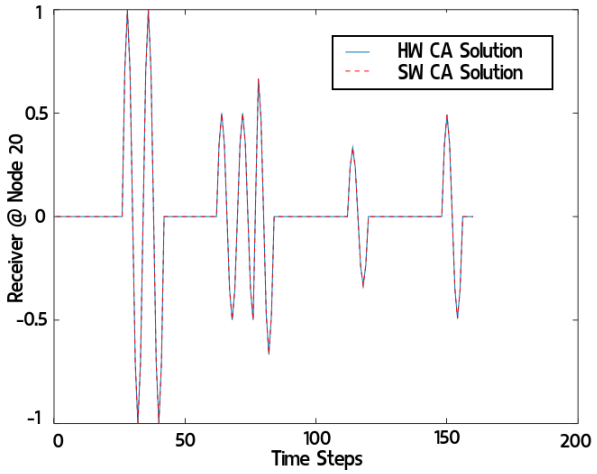


Fig. 7. Decoded results obtained from the "A" channel of Camphs1DB9G2 and its reference values.

In order to compare the differences between the *Camphs1DB9G2* and its reference model, the mean square error (MSE) criterion was adopted. The obtained CA solution, for both receivers, comes too close to the reference model with MSE in the tune of 1×10^{-13} or less, as shown in Table IV. The low margin of errors occurs because of the rounded applied data. Since the implemented model is vulnerable to the effects of digitization and numerical losses, it is not expected for an aleatory sampling data.

It is worth mentioning that all VHDL codes have been synthesized for the Zynq Ultrascale device of the ZCU104 development kit (chip xczu7evffvc1156-2-e). Resources utilization of the main architecture and each cell can be appreciated in Table V.

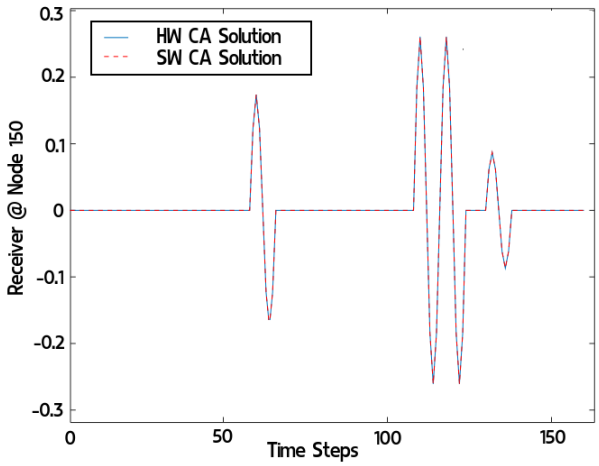


Fig. 8. Decoded results obtained from the "B" channel of Camphs1DB9G2 and its reference values.

TABLE IV
MEAN SQUARE ERRORS OBTAINED FROM *Camphs1DB9G2* IN BOTH RECEIVER CHANNELS.

Receptor	MSE
A	3.45×10^{-13}
B	1.53×10^{-14}

As mentioned before, the cell's core uses floating point multipliers and adders with 27 bits. It lead us to connect the cells outputs with their neighborhood, directly. Another important point it that there is any state machine (FSM) inside the mentioned CA. This operational characteristic improves the Camphs1D speed. Consequently, the systems scalability ends up being a function of the core amount that works in parallel.

The execution time of the proposed *Camphs1D*, and its possible artificial organisms, depends on the latency and throughput of the cells as well as on the clock frequency. Because of the latency of four clock cycles of the *GCA* and *HCA* cells, the achieved throughput of the cells at a clock frequency of 100 MHz is around 25 MFLOPS. The latency of a *Camphs1D* organism depends on its length and the position of the receivers; therefore, after the initial latency, each receiver can provide a valid output every four-clock cycles, achieving

TABLE V
RESUME OF RESOURCES CONSUMPTION FOR *Camphs1DB9G2*.

Architecture	LUTs (230400)	Registers (460800)	DSPs % (1728)
Main architecture	22422 (9.73 %)	21924 (4.76 %)	372 (21.53 %)
GCA cell	114 (0.049 %)	118 (0.026 %)	2 (0.116 %)
HCA cell	825 (0.358 %)	220 (0.048 %)	4 (0.231 %)
HCA cell (cource)	303 (0.132 %)	83 (0.018 %)	1 (0.058 %)

25 MFLOPS.

VII. Conclusion

This work presented a novel architecture for cellular automata (CA) systems that can be used to model any one-dimensional grid of cells that evolves over discrete time steps. The developed CA system, called *Camphs1D*, has shown itself to be flexible and scalable, taking advantage of regular structures, that can be easily integrated through a high-level modeling tool called *vCAgen*. The proposed *vCAgen* tool is able to provide the VHDL code of the CA system and makes use of custom precision floating-point operators. In particular, this tool was used to implement a case study of a one-dimensional CA composed of 185 cells, allowing plane-wave phenomena to be simulated. Additional advantages are regarding circuit's simplicity when compared to alternative solutions in software. See more benefices listed below.

- It is possible to consider on-off states for cells, which is a tough task in software, because of the numerical nature of EDO discretization.
- Cell's mode is easily handled, requiring only to apply external signals to control it. In contrast, EDO-based solutions need more effort and rigid conditions for consistency and convergence.
- CA systems seen to be more efficient for being involved to physical phenomena digitization. EDO solutions need mode complex mathematical structures and algorithms to perform in a similar way.

On the other hand, modeling CA systems may require an appropriate degree of creativity and observation to obtain consistent results. The engineer/researcher must be able to choose, in each situation, the best approach equilibrium between realism and solution cost.

The only pointed disadvantage is regarding the necessary resources, which are highly costly. A trade-off between hardware acceleration and cost must be considered for each case application. Aiming to a far equilibrium, future steps go to the use of integrated hardware and software processors. Thus, it is expected to have DDR memories and a software processor, working to expand the CA system modeling capability.

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